

Tide retrospective: the gradients agree too

a second interpolation for the Boltzmann weights

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Abstract

For smooth nonnegative losses whose Laplace families agree beyond all orders on smooth compactly supported tests, the gradients of the Boltzmann weights agree uniformly on compacts beyond all orders: for every compact K and every N there is C with $\|D(e^{-tL_2} - e^{-tL_1})(x)\| \leq C t^{-N}$ for all $x \in K$ and all large t , and $D(e^{-tL_2} - e^{-tL_1})(x)v = o(t^{-\infty})$ at every point in every direction. The input is the uniform agreement of the weights from the previous tide; the new ingredients are a Hessian bound $\|D^2 e^{-tL}\| \leq t^2(\|DL\|^2 + \|D^2 L\|)$ and a deterministic interpolation lemma: if $|h| \leq A$ and $\|D^2 h\| \leq M$ on the closed unit ball about x , then $\|Dh(x)\| \leq 2A/s + Ms$ for every $0 < s \leq 1$, by the affine-error mean value inequality along $x + sv$. With $s = t^{-(N+3)}$ and the weight bound at exponent $2N+3$ the theorem follows. Three hundred lines, two LSP rounds, no surprises.

1 Setting

The local-uniform tide (2026-09-19-18-30) proved $\sup_K |e^{-tL_2} - e^{-tL_1}| = O(t^{-N})$ for every N , and its consult corrected a remark in my notes: the derivatives of the weight difference are *also* uniformly beyond all orders, by a second interpolation using the Hessian. This tide formalises that first derivative. The deliberation (tide log 2026-09-19-19-40-tide-germbij-derivatives.md in tide-log/, consult in gpt_germbij_deriv_v1.md) fixed one endpoint slip (closed balls, not open ones, since $x + sv$ lies on the sphere of radius s), recommended the affine-error mean value lemma over a one-dimensional reduction, and asked that the interpolation lemma be stated without reference to exponentials. Both parties voted for the package and for stopping at the first derivative.

2 The thing formalised

Write $w_i = e^{-tL_i}$, $h_t = w_2 - w_1$, $K' = \overline{K_1}$ for the closed unit thickening.

(H) $Dw = w(-tDL)$ as a function, $D^2w = Dw \otimes (-tDL) + w(-tD^2L)$, and $\|D^2w(y)\| \leq t^2(\|DL(y)\|^2 + \|D^2L(y)\|)$ for $t \geq 1$. On a compact S : $\|D^2h_t\| \leq B t^2$ for $t \geq 1$.

- (I) If h is twice differentiable on $\overline{B}(x, 1)$ with $|h| \leq A$ and $\|D^2h\| \leq M$ there, then $\|Dh(x)\| \leq 2A/s + Ms$ for every $0 < s \leq 1$.
- (G) Under exact smooth-test agreement, for every compact K and every N there is C with $\|Dh_t(x)\| \leq Ct^{-N}$ for all $x \in K$ eventually; and $Dh_t(x)v = o(t^{-\infty})$ for all x, v .

The Lean signature of (G):

```
theorem eventually_uniform_norm_fderiv_exp_sub_le {L1 L2 : (iota -> R) -> R}
  (h1 : ContDiff R infy L1) (h2 : ContDiff R infy L2)
  (hL1 : forall w, 0 <= L1 w) (hL2 : forall w, 0 <= L2 w)
  (hexact : forall phi, ContDiff R infy phi -> HasCompactSupport phi ->
    SuperPoly fun t => I2 phi t - I1 phi t)
  {K : Set (iota -> R)} (hK : IsCompact K) :
  forall N : Nat, exists C : R, eventually t in atTop, forall x in K,
    ||fderiv R (fun w => exp (-(t * L2 w)) - exp (-(t * L1 w))) x||
    <= C * t ^ (-(N : R))
```

(integrals abbreviated). The other declarations:

- (H), the weight: `fderiv_expWeight_eq` and `hasFDerivAt_fderiv_expWeight`;
- (H), the bound: `norm_fderiv_fderiv_expWeight_le`;
- the difference: `fderiv_weightDiff_eq` and `hasFDerivAt_fderiv_weightDiff`;
- the compact bound: `exists_hessian_bound_on`;
- (I): `norm_fderiv_le_of_bounds`;
- the pointwise corollary `superPoly_fderiv_exp_sub_at`;
- a helper `norm_expWeight_smul_le` valid in any normed space.

3 Proof strategy

Hessian. The derivative field $y \mapsto w(y)(-tDL(y))$ is a scalar function times a vector field; `HasFDerivAt.smul` gives its derivative $w(y)(-tD^2L(y)) + (Dw(y)).\text{smulRight}(-tDL(y))$, whose norm is at most $t\|D^2L\| + (t\|DL\|)^2$ using $0 < w \leq 1$ and `norm_smulRight_apply`; for $t \geq 1$ this is at most $t^2(\|DL\|^2 + \|D^2L\|)$. The difference is bounded by the sum, and on a compact the continuous function $\|DL_1\|^2 + \|D^2L_1\| + \|DL_2\|^2 + \|D^2L_2\|$ is bounded (`ContDiff.continuous_fderiv` twice, the second through `contDiff_infty_iff_fderiv`).

Interpolation. For a unit vector v and $0 < s \leq 1$: the mean value inequality for Dh on $\overline{B}(x, 1)$ gives $\|Dh(z) - Dh(x)\| \leq Ms$ on $\overline{B}(x, s)$; the affine-error mean value inequality `Convex.norm_image_sub_le_of_norm_fderiv_le'` on $\overline{B}(x, s)$ with $\varphi = Dh(x)$ then gives $|h(x + sv) - h(x) - sDh(x)v| \leq Ms \cdot s$. Hence $s|Dh(x)v| \leq 2A + Ms^2$, and `ContinuousLinearMap.opNorm_le_of_unit_norm` turns the directional bound into the operator-norm bound.

Assembly. On K' take $A = C_1 t^{-(2N+3)}$ from the uniform weight theorem and $M = Bt^2$ from the Hessian bound; with $s = t^{-(N+3)}$, $2A/s = 2C_1 t^{-N}$ and $Ms = Bt^{-(N+1)} \leq Bt^{-N}$ for $t \geq 1$. The pointwise corollary takes $K = \{x\}$ and $|Dh(x)v| \leq \|Dh(x)\| \|v\|$.

4 Roadblocks and resolutions

Two LSP rounds, both trivial: an `nlinarith` that needed the intermediate fact $t\|D^2L\| \leq t^2\|D^2L\|$ spelled out, and a rewrite chain where `Real.norm_eq_abs` fired on the left-hand norm before `abs_of_pos` could reach $\|s\|$; `Real.norm_of_nonneg` directly on $\|s\|$ fixed it. The lambda-versus-Pi discipline of the previous tides (name every derivative fact with its function type before rewriting with `.fderiv`) was applied from the start and caused no round.

The consult, before writing, caught the endpoint:

GPT-6 Astra: Endpoint correction: when $s = 1$, $x + sv$ need not belong to `ball x 1`. Use $[x, x + sv] \subseteq \text{closedBall}(x, 1) \subseteq \text{cthickening}(1, K)$. Likewise, $x + sv \notin \text{ball}(x, s)$ for unit v ; use `closedBall x s` in the Taylor step.

and prescribed the tool:

GPT-6 Astra: Apply the affine-error version of the convex mean-value estimate on `closedBall x s`, with $\varphi = Dh(x)$, $C = Ms$. It gives $|h(x + sv) - h(x) - Dh(x)(sv)| \leq Ms\|sv\| = Ms^2$. [...] I would package the resulting analytic lemma independently of exponentials.

5 What was Mathlib, what was new

Mathlib, all lookup:

- derivatives: `HasFDerivAt.smul`, `HasFDerivAt.const_smul`, `HasFDerivAt.neg`, `HasFDerivAt.sub`;
- regularity: `contDiff_infty_iff_fderiv`;
- continuity: `ContDiff.continuous_fderiv` and `ContDiff.differentiable`;
- operator norms: `ContinuousLinearMap.opNorm_le_of_unit_norm`;
- `ContinuousLinearMap.norm_smulRight_apply`;
- `ContinuousLinearMap.le_opNorm`;
- mean value: `Convex.norm_image_sub_le_of_norm_fderiv_le` and its affine-error variant;
- thickenings: `IsCompact.cthickening` and `Metric.mem_cthickening_of_dist_le`;
- bounds: `IsCompact.exists_bound_of_continuousOn`.

Seabed:

- `eventually_uniform_abs_exp_sub_le`;

- `hasFDerivAt_expWeight` and `expWeight_le_one`;
- `superPoly_of_forall_eventually_le`.

New: the Hessian identity and bound, the compact Hessian bound for the difference, the interpolation lemma, the uniform theorem and the pointwise corollary. About 300 lines.

6 Lessons

- For `CLAUDE.md`: when a rewrite chain has to normalise both a real norm on the left and $\|s\|$ on the right, use `Real.norm_of_nonneg` for the specific scalar rather than the generic `Real.norm_eq_abs`, which fires on the first norm it sees.
- For the tide skill: two consecutive interpolation tides ($L^1 \rightarrow L^\infty$, $L^\infty \rightarrow \text{gradient}$) shared their shape; the second cost half the first. Recording the mechanism as a template in the retrospective paid off within the hour.
- For the paper: exact agreement beyond all orders now holds for the weights and their gradients, uniformly on compacts. The note can state the normalized-case closure at the level of the densities.

7 Follow-ups

- *All orders*. Induction on k : $\|D^{k+1}h_t\| \leq B_k t^{k+1}$ on compacts (polynomial growth of all derivatives of e^{-tL}), then the interpolation lemma applied to $D^{k-1}h_t$ with Banach codomain. Needs the interpolation lemma restated for vector-valued h (the proof is unchanged) and bookkeeping of enlarged compacts.
- *Normalized densities and their gradients* uniformly on compacts, once the anchor bounds the partition values below.

Acknowledgements

GPT-6 Astra stated the result in the previous consult, corrected the endpoint, and named the affine-error mean value lemma. The seabed is the day's germbij tides.